

Rising Bandits for Best Athlete Identification

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Introduction

Rising and rotting rested bandits are a form of non-stationary bandit problem in which the reward function (payoff + noise) for an arm is a function of the number of times it has been pulled. Rotting bandits is a subset of this where the rewards follow a non-increasing function, and rising bandits is a different subset where the rewards are non-decreasing. Typically, these underlying reward functions are fixed and unknown. These problems can be used to model a number of real-world tasks. Heidari et al. relate that rising bandits can be used to model scenarios where workers learn/train at a task that is challenging, but gets easier with time. Meanwhile, rotting bandits can be used to model a task where workers lose interest (something tedious, for example). Rising bandits has been studied for its use in online model selection, hyperparameter optimization, and best arm identifications, among the applications of which is picking the best athlete out of a group. In this paper, I aim to survey the current state of the literature on rising bandits, with a small foray into rotting bandits as it is a similar problem. Additionally, I will relate this area of research to the development of baseball projection systems, through the introduction of a novel dataset of real-world plate appearances taken by MLB players who debuted starting in 2020.

Papers Surveyed

The most relevant papers that I have discovered on rising bandits are as follows:

- [Heidari et al. 2016](#): **Tight Policy Regret Bounds for Improving and Decaying Bandits**
- [Seznec et al. 2019](#): **Rotting Bandits Are Not Harder Than Stochastic Ones**
- [Li et al. 2020](#): **Efficient Automatic CASH via Rising Bandits**
- [Cella et al. 2021](#): **Best Model Identification: A Rested Bandit Formulation**
- [Metelli et al. 2022](#): **Stochastic Rising Bandits**
- [Mussi et al. 2024](#): **Best Arm Identification for Stochastic Rising Bandits**
- [Amichay et al. 2025](#): **Rising Rested MAB with Linear Drift**

Background

The different papers make various assumptions about the reward functions involved, and Heidari et al. 2016 and Li et al. 2020 consider algorithms that work with deterministic reward. Sez nec, Cella, Metelli, Mussi, and Amichay, however, introduce stochastic algorithms with proven regret bounds. Noise is typically assumed to be 1-sub Gaussian in these stochastic algorithms.

Heidari demonstrates that in rising bandits any non-concave reward function is not learnable, by proving a lower bound in the order of $O(n)$. Because of this fact, virtually all papers surrounding rising bandits make the assumption that rewards are concave (or that losses are convex). Amichay is able to prove additional properties when the concave reward function is linear, which makes constructing an estimator reasonably easy. Other authors, including Cella, consider a general function class with known parameters which can be learned.

Several different measures of regret are used to evaluate rising bandits algorithms, but typically not external regret, which is the standard optimal-minus-chosen at every timestep. External regret is not ideal because the optimal reward at a time step depends on the arms chosen through the entire history up to that point, so this would be computing the optimal rewards only for the sequence of reward functions that are generated by the player making potentially non-optimal choices. This is obviously not a true optimum.

There is a notion of *policy regret* used by many of the papers surveyed, which compares the best possible policy’s entire history to the history chosen by the actual policy. However, this is extremely difficult to compute without brute force optimization. Amichay’s linear drift setting is useful in that the authors prove that the optimal policy is always taking the same arm every time, meaning that the policy regret is equivalent to the standard static/fixed-arm result used in adversarial bandits. The best arm identification papers (Li, Cella, Mussi), define a *simple regret* which compares the reward of playing the optimal arm at every step in the horizon to the reward of playing the arm selected as “best” at every step in the horizon. Mussi also defines upper and lower bounds on error probability, although the upper bound is not sublinear.

Results

Rotting bandits, despite being a very similar setting on the surface, has a very different set of solutions. Heidari finds that the deterministic rotting bandits problem can be solved with zero/constant regret with a simple greedy algorithm: pick the arm with the highest last reward. This works because the best rewards come at the start of the run, whereas even with deterministic rising bandits, it is difficult to know if an arm which is not currently optimal will eventually become so.

All of the papers surveyed attempt to solve rising bandit problems with a variant of explore-then-commit, successive elimination, or UCB. For the UCB-based algorithms, the key factor is that due to the concave nature of the reward, the upper bound on reward is always a linear increase until the horizon, and the lower bound is a flatline to the horizon. These are used as optimistic and pessimistic estimators.

Amichay proves a tight $O(T^{4/5}K^{3/5})$ bound for rising bandits with linear drift. However, this does require a known horizon.

Online Model Selection and Best Arm Identification

Li et al. 2020 was the first paper to use rising bandits to study hyperparameter optimization, and proves an instance-specific regret that scales with the amount of trials it takes to distinguish the

optimal arm (this is a common type of bound). Li works with a deterministic reward function, but Cella et al. 2021 works towards best arm identification by learning the parameters of a known function class, including an Explore-then-Commit algorithm along with successive elimination.

As $t \rightarrow \infty$, rising bandit best arm/model identification approaches stationary best arm/model identification. Mussi et al. 2024 extends Li and Cella’s analysis with UCB and successive elimination algorithms for best arm identification.

Contributions

Mussi et al. mentions athlete training as a scenario that can be modeled with rising bandits:

Consider the role of a professional trainer for a team, having several athletes (i.e., our arms) to train in order to increase their performances. The final goal is to select a single athlete to represent the team in a competition. The performances of athletes increase when the trainer properly follows them. However, a trainer can follow just one athlete at a time. The trainer can be modeled as an agent performing best arm (athlete) identification, and the athletes represent the arms that increase their payoffs (i.e., performance) when pulled (i.e., when the trainer follows them).

Baseball lineup selection could be modeled in a similar way. To work towards this, I constructed a dataset of baseball plate-appearance outcomes, sourced from Retrosheet’s play-by-play and biographical data. This data, found in `youths.csv` in the attached GitHub repo, consists of ostensibly Bernoulli rewards of whether a hitter got on base. I include only players who debuted starting in 2020 and who have at least 300 plate appearances. This allows for their “underlying reward function” to be estimated with reasonable closeness.

Essentially, this allows the first 1–5 years of a player’s career to be modeled as a rising bandits problem with an increasing underlying reward function. Baseball is well-suited to this as players typically hit their prime 5–6 years after entering the league, meaning that they regularly improve for longer than athletes in other sports do.

This dataset works in tandem with best arm identification algorithms, and I have implemented Mussi’s successive rejection algorithm with this dataset (using the Mussi paper’s provided code), producing the following results:

Code can be found here: <https://github.com/samc5/rising-bandits>

Open Questions

I was hoping to have time to study contextual rising bandits, which seems to be an omission in the current literature. In terms of baseball, context could be the opponent, or even more relevant, the handedness of the batter and the opposing pitcher.

Regret Analysis

Below are two plots.

Regret vs Number of Arms for Different Horizons

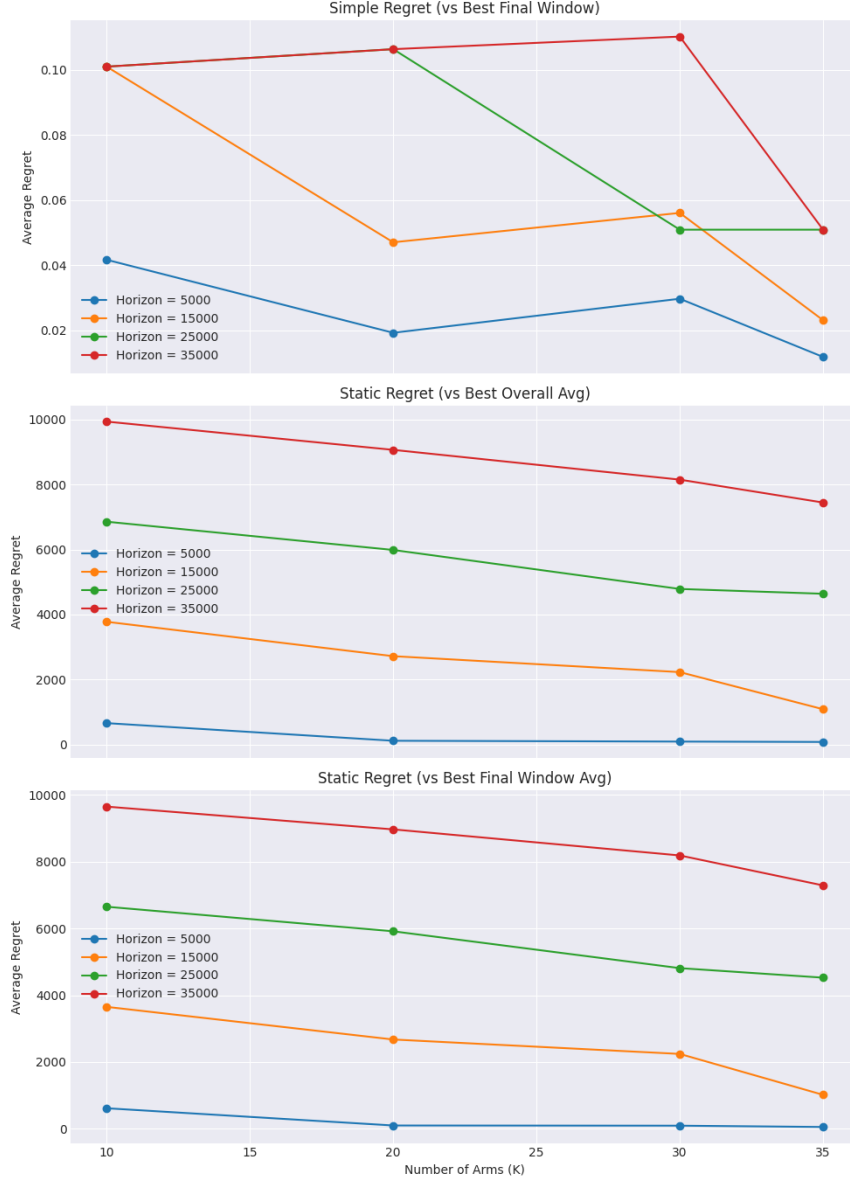


Figure 1: Horizon vs Regret. Displays a linear regret for all regret metrics except for static regret. This indicates that the model is not working properly on the dataset, although the estimates seem to be a reasonable approximation of on-base percentage. The simple regret plateaus at higher levels, suggesting that there is some learning happening here.

Regret vs Horizon for Different Numbers of Arms

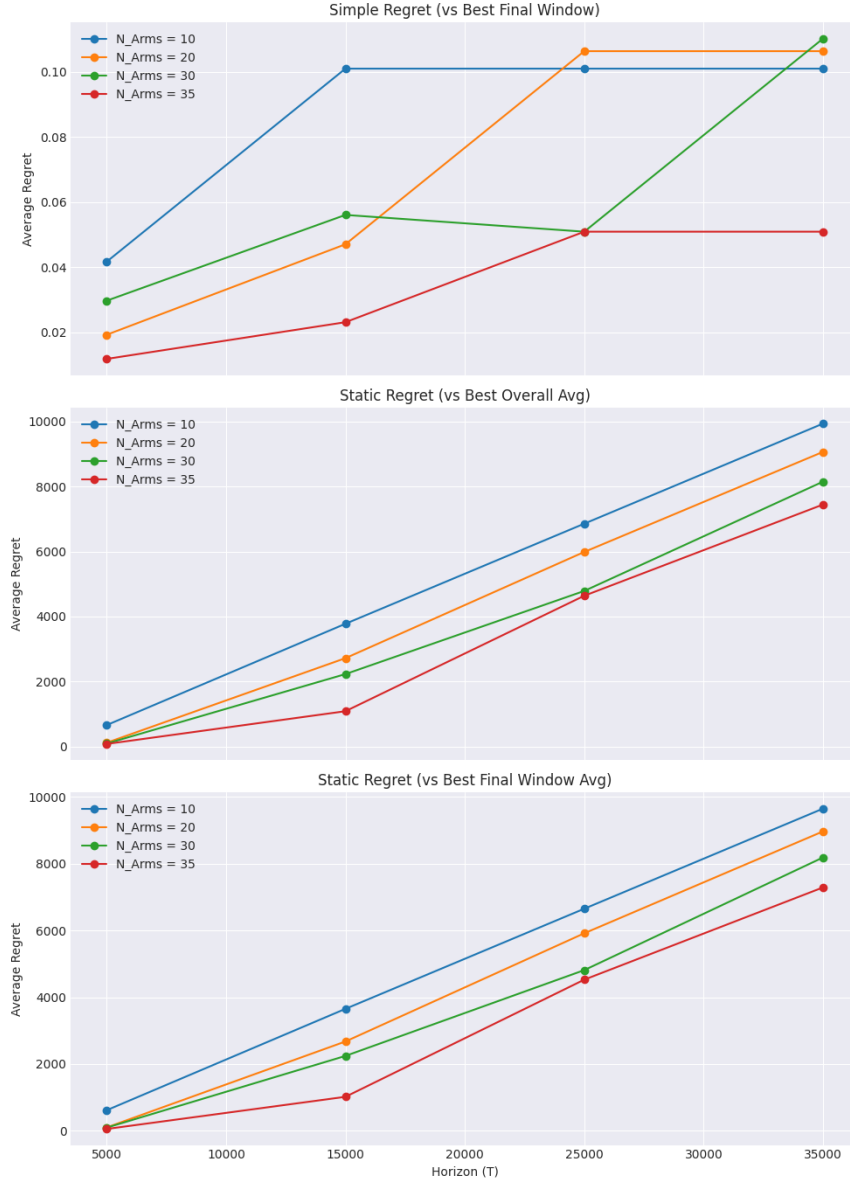


Figure 2: This graph mostly shows that more arms leads to a greater distribution among them and therefore lower total regret for the same horizon.